Spatial representations emerge from temporal event credit in a hippocampus-inspired navigation agent without spatial supervision

Can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, exploring whether spatial organization can arise from the temporal structure of sensory experience and self-generated behavior, without position labels, maps, or spatial rewards.

Our model combines dentate gyrus (DG)-like sparse sensory events with CA3-like sequential memory. Temporal relationships between encountered DG events provide two complementary learning signals: a **"push"** that reinforces DG activity more strongly after longer intervals between events, and a **"pull"** that rewards shorter transitions between sensory events or, in goal-conditioned variants, faster arrival at internally defined DG targets. Together, these pressures aim to make temporal separation a more informative proxy for navigational distance. The agent learns from egocentric visual experience and intrinsic motivations, without explicit information about its physical location or spatial distances.

During exploration of a continuous visual environment, spatially informative DG activity emerges. To isolate the contribution of temporal information, we compare temporally weighted DG feedback against mean-matched constant feedback, evaluating the resulting representations on identical sensory observations. Temporal feedback produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks across five paired training seeds. These improvements extend to downstream decoder representations, where spatial information also increases consistently. In contrast, temporally weighting the policy's target-hit reward does not consistently improve exploration or spatial representation quality compared with constant hit rewards.

The agent's behavior further shapes spatial coding. Even with network weights frozen, the learned policy produces more spatially informative downstream activity than random actions, demonstrating that spatial coding depends not only on the learned network but also on how the agent interacts with its environment.

Together, our results demonstrate how spatial coding can emerge from the interaction of temporal learning, sequential memory, and self-generated behavior, without explicitly optimizing for spatial information.

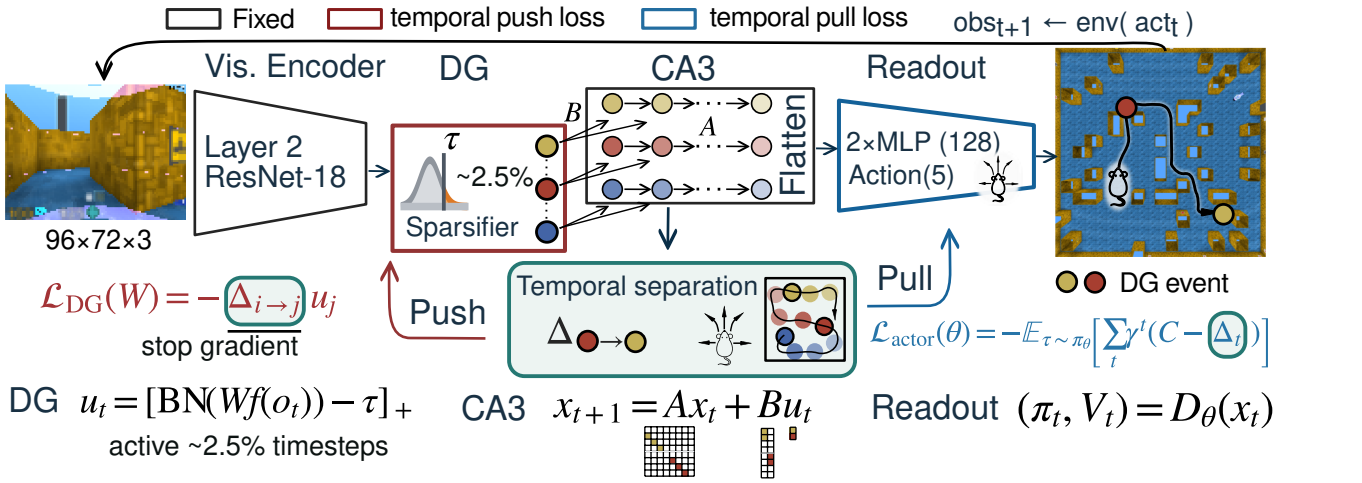


Figure 1: Push–pull architecture. A fixed pretrained visual trunk and trainable sparse dentate gyrus (DG) layer convert egocentric images into events. Fixed CA3-like memory stores event sequences; a learned readout controls actions and also receives egocentric depth. Red marks DG credit weighted by time since the previous eligible event (push). Blue marks policy reward for short event-to-event transitions or, in goal-conditioned agents, internal DG target hits (pull). C01 explores directly; C05 uses a DG target; C15 adds frontier-based target choice. No spatial coordinates, map, or spatial reward is supplied during learning.

Three tests. We measure spatial coding along evolving trajectories, change actions with network weights fixed, and replay identical observations through separately trained networks (Fig. 2). These comparisons distinguish the emergence of a spatial code during learning, its expression under different behavior, and the contribution of temporal DG credit to the learned network. Common replay removes evaluation-sampling differences, although training paths may still matter.

Spatial emergence and temporal push. DG coding rises online as the agent explores (Fig. 2A). With the same 10,001 observations replayed through each trained network, temporal DG credit improves the baseline DG spatial score in all five matched seeds (mean +0.0575), reduces map overlap, and broadens the area covered by eligible field peaks (mean +0.109; Fig. 2C,D). Seed 8 illustrates the largest peak-area increase among these pairs, while panel D shows the full distribution. The improvement reaches the downstream readout: decoder-1 spatial information rises in all five baseline pairs (mean +0.680 bits per activation; Fig. 2E). Goal-conditioned encoder comparisons largely point the same way, although the frontier-goal condition currently has only three matched pairs.

Behavior shapes spatial coding. With each network frozen, its learned policy elicits more spatially informative decoder-1 activity and more reliable spatial maps than random actions in all 38 matched models (Fig. 2B). Thus, behavior changes the spatial signal available from a fixed network. This test identifies an effect of the agent's interaction with the environment at evaluation time; it does not by itself show how policy changes shaped the network during training.

Temporal pull. Adding temporal weighting to the goal-conditioned policy's target-hit reward has mixed effects on coverage and DG fields (Fig. 2F). Both worker bonuses reward detector hits, and actor–critic discounting already favors earlier hits, which may weaken the additional effect of explicit temporal weighting. A separate longer-credit intervention changes PPO discount and GAE lambda jointly: frontier-policy coverage rises while spatial information falls on separate policy rollouts. Because those rollouts sample different experiences, this observation does not isolate a fixed-input change in DG coding. Detector hits also do not establish physical goal arrival; matched-command and physical-arrival analyses are needed to assess goal control.

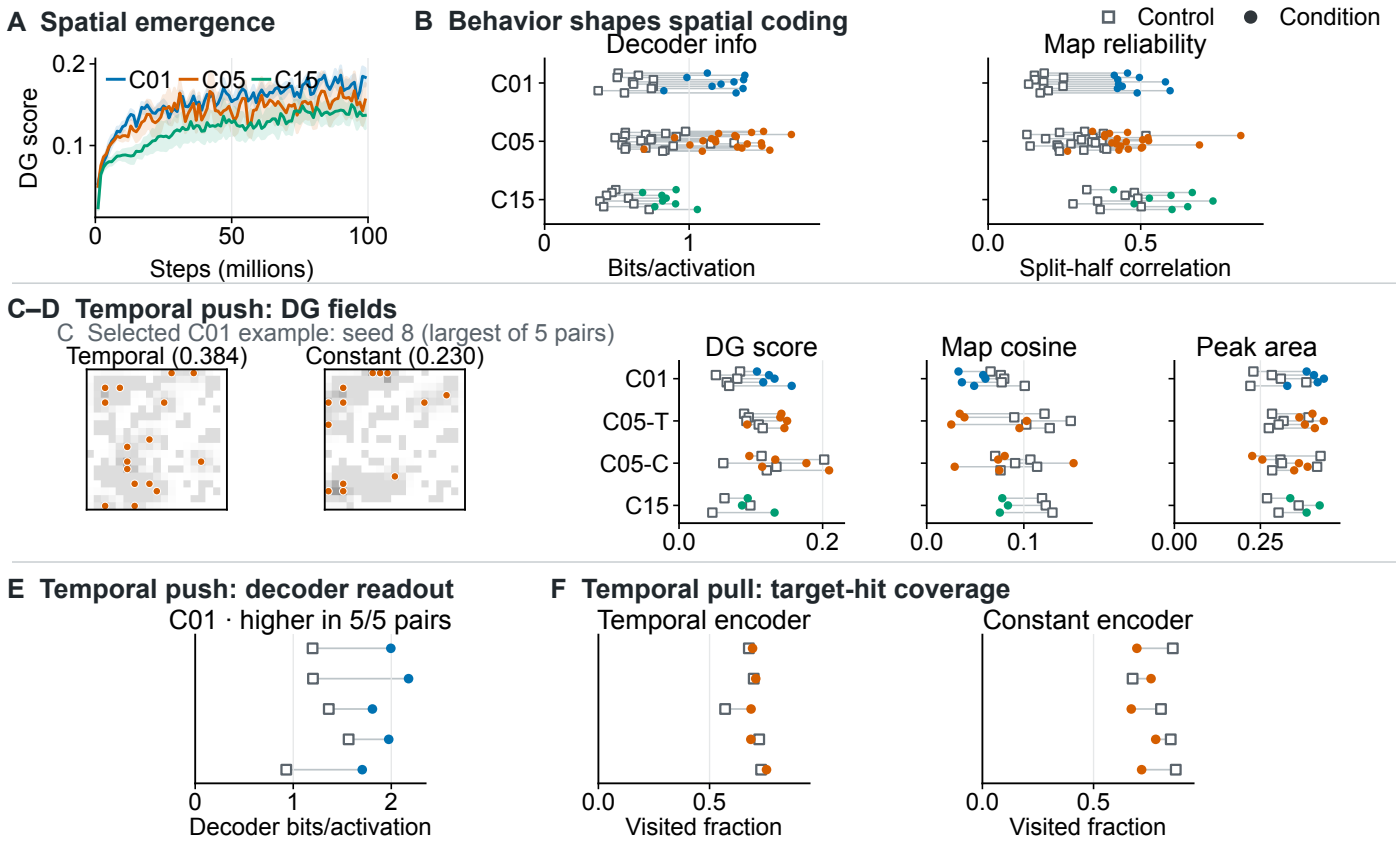


Figure 2: Temporal credit, behavior, and spatial coding. Blue, orange, and green denote baseline (C01), goal-conditioned (C05), and frontier-goal (C15) agents. Open gray squares are controls, filled circles the tested condition, and gray lines connect matched values. **A,** Online, amplitude-weighted DG spatial score on each policy's latest 10,000 observations: five-seed mean ±1 s.e.m. All curves use temporal DG credit; C05/C15 also use temporal worker hit credit. **B,** Frozen-model decoder-1 spatial information and split-half map correlation under random actions versus the learned policy. Each pair represents one of 38 terminal models, averaged over two 50,000-decision evaluations. Information is in bits per activation on equally weighted shared bins and jointly eligible units; correlation uses bins visited at least ten times in each episode-disjoint half of both policies. Both readouts favor the learned policy in all 38 models at this threshold. **C,** Baseline seed 8, selected for the largest peak-area difference among five paired seeds: temporal 0.384 versus constant 0.230 on the same 10,001 observations. Dots are DG-unit peaks after a 0.25% activity cutoff; grayscale shows Gaussian-smoothed peak density, and light gray marks unvisited bins. Effective peak area is exponential entropy of equal-mass peak density divided by 242 visited bins (100-unit Gaussian width). **D,** Temporal DG credit versus mean-matched constant control on the identical observation sequence at terminal checkpoints. Rows hold policy and worker bonus fixed: baseline (n = 5 seed pairs), C05 with temporal or constant hit bonus (Goal-T/C; n = 5 each), and C15 (n = 3). DG score is amplitude-weighted; lower mean cosine among active-unit maps indicates less overlap; peak area is defined in C. **E,** Baseline decoder-1 information per activation on that shared sequence: temporal DG credit is higher in all five seed pairs (mean gain 0.680 bits per activation). **F,** C05 online visited-cell fraction averaged over 90–99 million training frames: temporal versus mean-matched constant worker hit bonus, with temporal or constant DG credit held fixed (n = 5 pairs each). Mean temporal-minus-constant differences are +0.026 and −0.087; this is coverage, not executed goal control.